AIML LAB MANUAL

Electric Vehicle (EV) Charging Station Route Planning

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Experiment Title: Comparative Study of Search Algorithms for Electric Vehicle Route Planning

Application: AI-based electric vehicle (EV) route planning and optimisation under battery and mandatory charging-station constraints.

1. AIM

To design and implement multiple search algorithms — uninformed, informed, local, and constraint-satisfaction based — in Python to plan a feasible route for an electric vehicle (EV) that must visit a set of mandatory charging stations while never letting its battery fall below a minimum safe level, and to compare the algorithms on route quality, execution time, memory usage and search effort.

2. THEORY

An intelligent agent perceives its environment, reasons about the current state, and selects an action accordingly. In this experiment, the agent is an electric vehicle that must travel from a fixed START location to a fixed DESTINATION while passing through a specified subset of charging stations (CS2, CS5 and CS6) so as to never run out of charge.

The environment consists of ten locations — START, eight charging stations (CS1–CS8) and DEST — placed on a 2-D plane. The distance between any two locations is the Euclidean distance between their coordinates, and travelling a distance consumes battery at a fixed rate (0.18 kWh/km). Reaching one of the three mandatory charging stations recharges the battery by a fixed amount, up to the battery capacity. A route is feasible only if the battery never drops below the minimum threshold (5 kWh) and all three required charging stations are visited before reaching DEST.

A state for the search is therefore represented as the triple (current location, set of required stations already visited, current battery level). Nine different search strategies were implemented and compared on this state space:

• Uninformed search — Breadth-First Search (BFS), Depth-First Search (DFS) and Uniform Cost Search (UCS), which explore the state space without any domain-specific estimate of the remaining distance.

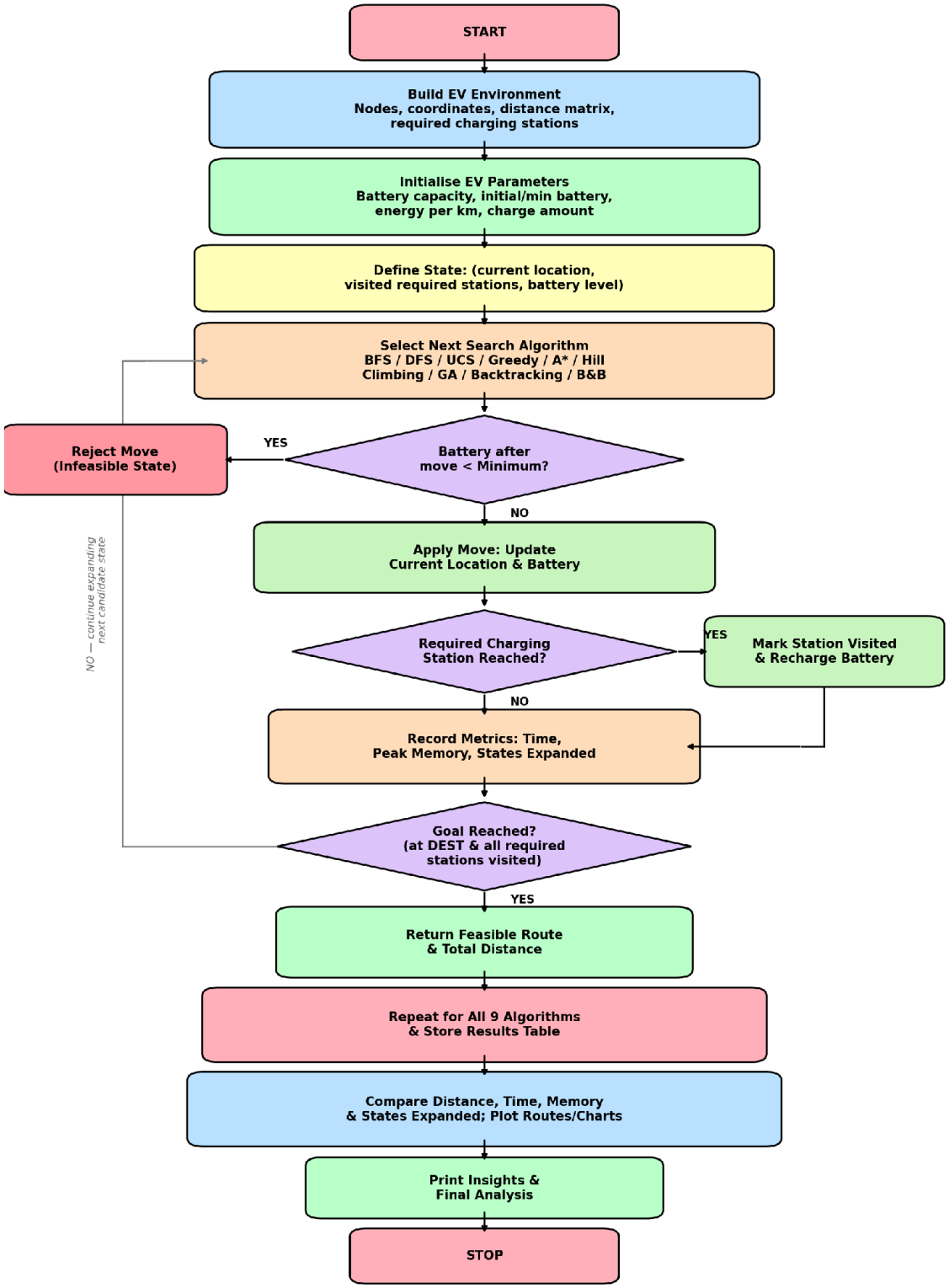
• Informed search — Greedy Best-First Search, which expands the state that appears closest to the goal according to a heuristic, and A* Search, which combines the accumulated route cost with the heuristic estimate.

• Local search — Hill Climbing, which iteratively improves a feasible route using random restarts, and a Genetic Algorithm, which evolves a population of candidate routes through selection, crossover and mutation.

• Constraint Satisfaction — Backtracking, which systematically assigns the order of required stations while respecting the battery constraint, and Branch and Bound, which additionally prunes branches whose lower-bound cost cannot beat the current best solution.

The heuristic function used by Greedy Best-First Search and A* estimates the remaining distance as the sum of the distance to the nearest unvisited required station and the distance from that station to the destination.

3. FLOWCHART


START
Build EV Environment
Nodes, coordinates, distance matrix,
required charging stations
Initialise EV Parameters
Battery capacity, initial/min battery,
energy per km, charge amount
Define State: (current location,
visited required stations, battery level)
Select Next Search Algorithm
BFS / DFS / UCS / Greedy / A* / Hill
Climbing / GA / Backtracking / B&B
Battery after
move < Minimum?
YES
Reject Move
(Infeasible State)
NO — continue expanding
next candidate state
NO
Apply Move: Update
Current Location & Battery
Required Charging
Station Reached?
YES
Mark Station Visited
& Recharge Battery
NO
Record Metrics: Time,
Peak Memory, States Expanded
Goal Reached?
(at DEST & all required
stations visited)
YES
Return Feasible Route
& Total Distance
Repeat for All 9 Algorithms
& Store Results Table
Compare Distance, Time, Memory
& States Expanded; Plot Routes/Charts
Print Insights &
Final Analysis
STOP

4. OBSERVATION

EV Environment Parameters

Parameter	Value
Nodes	START, CS1–CS8, DEST
Required charging stations	CS2, CS5, CS6
Battery capacity	60.0 kWh
Initial battery	45.0 kWh
Minimum battery	5.0 kWh
Energy consumption	0.18 kWh/km
Charge added per station	30.0 kWh

Algorithm Comparison Results

Algorithm	Distance (km)	Time (s)	Memory (MB)	States Exp.	Iterations	Generations	Restarts
BFS	129.57	0.0826	1.350	500	-	-	-
DFS	113013.61	0.7751	136.459	2622	-	-	-
Uniform Cost Search	129.57	0.0130	0.236	86	-	-	-
Greedy Best-First	129.57	0.5325	3.756	1718	-	-	-
A*	129.57	0.0032	0.025	12	-	-	-
Hill Climbing	129.57	0.0069	0.001	-	40	-	16
Genetic Algorithm	129.57	1.3258	0.038	-	-	150	-
CSP Backtracking	129.57	0.0004	0.003	14	-	-	-
CSP Branch & Bound	129.57	0.0004	0.003	9	-	-	-

Best Results Summary

Metric	Algorithm
Shortest feasible route	BFS — 129.57 km
Fastest execution	CSP Branch & Bound — 0.000377 s
Lowest memory usage	Hill Climbing — 0.001 MB
Least search effort (states expanded)	CSP Branch & Bound — 9 states

5. CODE FILE LINK

COLLAB LINK -
https://colab.research.google.com/drive/1q58nnWMu1Org0Fy-hPsSdnJjtsQ4Z_ym?usp=sharing

6. RESULT ANALYSIS

1. EV Route Quality

The shortest feasible route observed in this experiment was produced by BFS, with a total travel distance of 129.57 km. The route satisfies the required charging-station and battery constraints defined for the EV problem. In fact, eight of the nine algorithms (all except DFS) converged to this same optimal distance of 129.57 km, confirming that this is the true shortest feasible route for the given environment.

2. Execution Time

The lowest measured execution time was obtained by CSP Branch & Bound, with an execution time of 0.000377 seconds, closely followed by CSP Backtracking. This result reflects the current dataset and implementation, and should not be interpreted as a universal ranking of the algorithms.

3. Memory Usage

The lowest measured peak memory consumption was observed for Hill Climbing, at approximately 0.001 MB. Memory usage depends on the number of states stored in memory and the implementation details of each search strategy — algorithms that keep large frontiers or visited-state sets (such as DFS) consume noticeably more memory.

4. Search Effort

Among the algorithms for which states expanded were measured, CSP Branch & Bound expanded the fewest states (9). Algorithms such as BFS and DFS explore a much larger number of EV states, because the state representation includes the current location, the set of visited charging stations and the battery level, which multiplies the size of the search space.

5. Uninformed Search

BFS, DFS and UCS do not use any domain-specific estimate of the remaining distance. BFS explores states level by level and found the optimal 129.57 km route while expanding 500 states; UCS reached the same result while expanding far fewer states (86) by prioritising accumulated route cost; DFS followed deep branches first and, in this configuration, produced a highly sub-optimal route (113013.61 km) while expanding the most states (2622).

6. Informed Search

Greedy Best-First Search uses only the heuristic estimate of the remaining route, while A* combines the accumulated route cost with the heuristic. For EV route planning, this heuristic information dramatically reduced the search effort: A* reached the optimal 129.57 km route while expanding only 12 states — the fewest among all uninformed/informed searches — by directing the search toward the required charging stations and the final destination.

7. Local Search

Hill Climbing attempts to improve a current feasible route by examining neighbouring orderings of the required stations, using random restarts to escape local optima; it found the optimal route within 40 iterations and 16 restarts while using negligible memory. The Genetic Algorithm instead maintains a population of candidate routes and improves them over generations through selection, crossover and mutation, converging to the optimal route within 150 generations. Both local-search methods found good solutions without exhaustively exploring every possible route.

8. Constraint Satisfaction

The CSP approaches explicitly enforce the required charging-station and battery constraints while assigning the visiting order. Backtracking systematically tries candidate orderings and reached the optimal route while expanding only 14 states; Branch & Bound additionally prunes branches whose lower-bound cost cannot improve upon the current best solution, and thereby achieved both the fastest execution time and the fewest states expanded (9) of all nine algorithms.

9. EV-Specific Conclusion

Unlike a conventional Travelling Salesman Problem, this EV routing problem is not based on distance alone. A route is feasible only when the EV has sufficient battery to travel between consecutive locations and all required charging stations are visited before reaching the destination. Therefore, battery capacity, energy consumption, charging opportunities, route distance and search effort all jointly influence which algorithm is best suited to a given deployment scenario — e.g. CSP Branch & Bound for speed and low memory on embedded systems, versus A* when a strong heuristic is available and optimality with minimal expansion is desired.

7. RESULT

Nine search algorithms — BFS, DFS, Uniform Cost Search, Greedy Best-First Search, A*, Hill Climbing, a Genetic Algorithm, CSP Backtracking and CSP Branch & Bound — were successfully implemented in Python to solve the electric-vehicle charging-station route-planning problem, and all except DFS converged to the optimal feasible route of 129.57 km, visiting the mandatory charging stations CS2, CS5 and CS6 without the battery ever falling below the minimum safe level. Among these, CSP Branch & Bound achieved the best combination of execution time and search effort (0.000377 s, 9 states expanded), while Hill Climbing used the least memory (0.001 MB).

8. CONCLUSION

This experiment demonstrates the working of an intelligent agent solving a constrained route-planning problem in a simulated electric-vehicle environment. By modelling the state as (location, visited charging stations, battery level) and comparing uninformed, informed, local, and constraint-satisfaction search strategies side by side, it is possible to observe how heuristic guidance (A*) and explicit constraint pruning (Branch & Bound) can drastically reduce search effort compared to purely uninformed exploration (BFS, DFS), while local-search methods (Hill Climbing, Genetic Algorithm) offer an efficient alternative that finds near-optimal or optimal routes without exhaustive search. Incorporating battery capacity, energy consumption and mandatory charging stops makes the simulation considerably more realistic than a plain shortest-path or TSP formulation, and better suited to real-world EV route-planning applications.